
QT 406: Introduction to Quantum Communication

Course Code: QT 406

Course Name: Introduction to Quantum Communication

L-T-P-C: 3-0-0-3

Prerequisites: Engineering Mathematics (Linear Algebra, Complex algebra, basics of 2nd of ODEs and initial value problems, 2nd order PDEs and boundary value problems, Probability and Statistics, Random variables). Maxwell's equations and EM theory at the level of the core physics syllabus from AICTE model

Intended for: UG/PG/PhD

Distribution: Elective PG/Elective UG

Semester: Odd/Even

Approval: 47th Senate

Preamble

Students of this course learn (i) The basics of EM theory, (ii) The basics of photodetection, (iii) The basics of information theory, (iv) The central ideas in quantum communications

Course Content and syllabus

- **Basics of Polarization optics**
 - Quarter and half-wave plates
 - Polarizing beam splitters
- **Basics of linear and square-law detectors**
 - Quadrature amplitude modulation
 - Heterodyne and Homodyne demodulation and linear detectors
 - Intensity measurements and square law detectors
 - Photomultipliers, Avalanche Photo diodes
- **Digital communication – information theory (basics)**
 - Information entropy
 - Noiseless channel encoding
 - Noisy channel encoding
- **No cloning theorem**
- **Quantum Memories**
- **Quantum repeaters**
- **Entanglement and Bell Theorems**
- **Bell Measurements and Tests**
- **Quantum Teleportation protocol**

- **Quantum Dense coding**
- **Quantum Key Distribution protocols**
 - BB84
 - E91
 - BBM92
 - B92
 - COW
 - DPS
- **Quantum Networks and Quantum Internet**
- **Survey of Hardware implementations**
 - Free space communications
 - Satellite based communications
 - Fibre optics-based communications

Course References

1. *Quantum computation and quantum information* – Nielsen and Chuang Cambridge University Press, Cambridge (2010)
2. A Pathak, *Elements of Quantum Computation and Quantum Communication*, Boca Raton, CRC Press (2015)